

# Tests for a parallelogram

The properties run backwards: three conditions, any one of which is enough to guarantee that a quadrilateral is a parallelogram.

LESSON 6

1 × 40 MIN

GEOMETRY 8, TEMA 3

STAGE 9 · BEYOND

LESSON CLOCK

4'

14'

7'

13'

2'

|             |        |
|-------------|--------|
| Warm-up     | 4 min  |
| Explanation | 14 min |
| Interactive | 7 min  |
| Practice    | 13 min |
| Homework    | 2 min  |

LEARNING OBJECTIVES

- State the three tests for a parallelogram.
- Tell a property (given a parallelogram, deduce ...) from a test (given ..., deduce a parallelogram).
- Prove a quadrilateral is a parallelogram using the appropriate test.
- Recognise a condition that is *not* sufficient.

TEXTBOOK

|           |                          |
|-----------|--------------------------|
| National  | Tema 3, pp. 11–13        |
| Cambridge | Extension beyond Stage 9 |
| Workbook  | —                        |

01

## Explanation

### Property or test?

Last lesson every statement began “*in a parallelogram ...*”. This lesson every statement ends “*... therefore it is a parallelogram*”. Same facts, opposite direction — and getting the direction right is most of the marks in a proof.

| PROPERTY (TEMA 2)                          | TEST (TEMA 3)                                            |
|--------------------------------------------|----------------------------------------------------------|
| A parallelogram has opposite sides equal.  | Opposite sides equal $\implies$ parallelogram.           |
| A parallelogram has opposite angles equal. | Opposite angles equal $\implies$ parallelogram.          |
| Its diagonals bisect each other.           | Diagonals bisecting each other $\implies$ parallelogram. |

### Test 1 — two opposite sides equal and parallel

TEST 1

If two **opposite** sides of a quadrilateral are equal **and** parallel, it is a parallelogram.

**Proof.** Suppose  $AB = CD$  and  $AB \parallel CD$ . Draw the diagonal  $AC$ . Then  $\angle BAC = \angle DCA$  (alternate angles) and  $AC$  is common, so  $\triangle ABC \cong \triangle CDA$  by SAS. Hence  $\angle BCA = \angle DAC$ , and those are alternate angles for  $BC$  and  $AD$  — so  $BC \parallel AD$  and the definition is satisfied. ■

#### BOTH CONDITIONS, ON THE SAME PAIR

Equal *and* parallel, and it must be the *same* pair of sides. Two sides equal and a different pair parallel gives an isosceles trapezium, not a parallelogram.

### Test 2 — both pairs of opposite sides equal

#### TEST 2

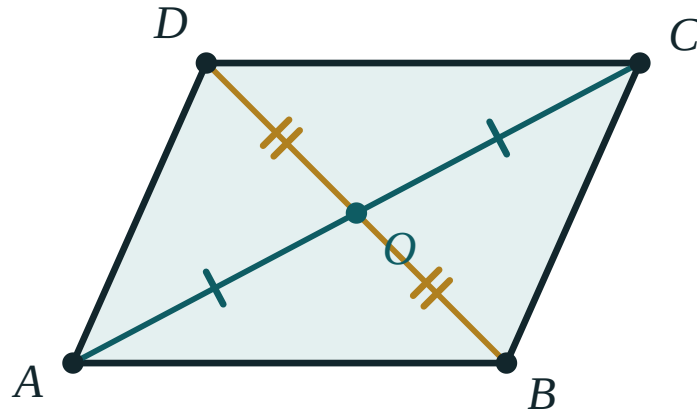
If  $AB = CD$  and  $BC = AD$ , the quadrilateral is a parallelogram.

**Proof.** The diagonal  $AC$  makes  $\triangle ABC \cong \triangle CDA$  by SSS. Then both pairs of alternate angles are equal, so both pairs of opposite sides are parallel. ■

### Test 3 — the diagonals bisect each other

#### TEST 3

If the diagonals of a quadrilateral bisect each other, it is a parallelogram.



*$AO = OC$  and  $BO = OD$  is enough on its own — the shape has to be a parallelogram.*

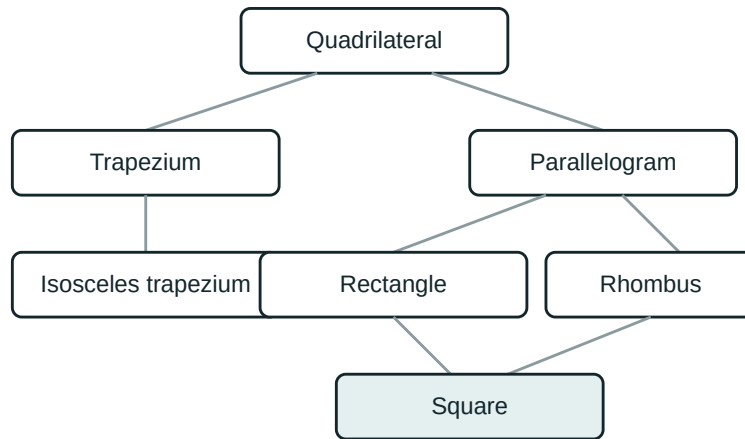
**Proof.** With  $AO = OC$ ,  $BO = OD$  and the vertically opposite angles at  $O$  equal,  $\triangle AOB \cong \triangle COD$  by SAS. So  $AB = CD$ , and the equal alternate angles give  $AB \parallel CD$ . Test 1 finishes the job. ■

#### A FOURTH TEST, SOMETIMES USEFUL

If both pairs of opposite **angles** are equal, the quadrilateral is a parallelogram. From  $\angle A = \angle C$  and  $\angle B = \angle D$  with total  $360^\circ$  we get  $\angle A + \angle B = 180^\circ$ , which makes the sides parallel as co-interior angles.

### Conditions that are not enough

- One pair of sides parallel only → a **trapezium**.
- One pair parallel and the *other* pair equal → could be an isosceles trapezium.
- Diagonals equal → could be an isosceles trapezium or a rectangle.
- Two adjacent sides equal → could be a kite.



Where each quadrilateral sits. An arrow means “is a special case of”.

## 02 Worked examples

**EXAMPLE 1** In quadrilateral  $ABCD$ ,  $AB = 7\text{ cm}$ ,  $CD = 7\text{ cm}$  and  $AB \parallel CD$ . Is  $ABCD$  a parallelogram?

① One pair of opposite sides is both equal and parallel.  
That is exactly Test 1.

② Test 1 applies to the same pair of sides  $AB$  and  $CD$  ✓  
Check that the pair really is opposite.

③ So  $ABCD$  is a parallelogram.

ANSWER

Yes — by Test 1.

**EXAMPLE 2** *The diagonals of  $ABCD$  meet at  $O$  with  $AO = OC = 5$  and  $BO = 4$ ,  $OD = 6$ . Is it a parallelogram?*

- ①  $AO = OC$  ✓  
One diagonal is bisected.

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- ②  $BO = 4 \neq 6 = OD$  ✗  
The other is not.

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- ③ Test 3 needs **both** diagonals bisected.  
One is not enough.

ANSWER

No — Test 3 fails.

**EXAMPLE 3**  *$M$  is the midpoint of both  $AC$  and  $BD$ . Prove  $ABCD$  is a parallelogram.*

- ①  $AM = MC$  and  $BM = MD$   
 $M$  is the midpoint of each diagonal.

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- ② So the diagonals bisect each other.  
Exactly the hypothesis of Test 3.

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- ③  $ABCD$  is a parallelogram.

ANSWER

Parallelogram, by Test 3.

### 03 Interactive model

Use the model to test each condition: which ones force the shape, and which leave it free?

## Is this enough to force a parallelogram?

One pair of opposite sides is equal **and** parallel.

① Draw diagonal

$AC$  . Alternate angles give

$\angle BAC$ .

=

$\angle DCA$

Uses

$AB$

$\parallel$

$CD$ .

②  $AB = CD$

,  $AC$

and

is  $\triangle ABC \cong \triangle CDA$

common,

so

by

SAS.

3 Then

$\angle BCA$ , so

=

$\angle DAC$

$BC$  .

//

$AD$

Both  
pairs  
are  
now  
parallel.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH              | ЎЗБЕКЧА        | РУССКИЙ              |
|----------------------|----------------|----------------------|
| Test (criterion)     | Alomat         | Признак              |
| Sufficient condition | Yetarli shart  | Достаточное условие  |
| Necessary condition  | Zarur shart    | Необходимое условие  |
| Converse statement   | Teskari tasdiq | Обратное утверждение |
| Counter-example      | Qarshi misol   | Контрпример          |
| Quadrilateral        | To'rtburchak   | Четырёхугольник      |
| Trapezium            | Trapetsiya     | Трапеция             |
| Kite                 | Deltoid        | Дельтоид             |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Which condition forces a quadrilateral to be a parallelogram?

its diagonals are equal

its diagonals bisect each other

two adjacent sides are equal

one pair of sides is parallel

$AB = CD$  and  $AB \parallel CD$  means the quadrilateral is:

a trapezium

a parallelogram

a kite

not determined

A quadrilateral with one pair of parallel sides only is:

a parallelogram

a trapezium

a rhombus

a rectangle



Both pairs of opposite angles equal means:

nothing in particular

it is a parallelogram

it is a rectangle

it is a rhombus

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $AB = CD$  and  $AB \parallel CD$ . Is  $ABCD$  a parallelogram?

ANSWER Yes — Test 1.

2.  $AB = CD$  and  $BC = AD$ . Is  $ABCD$  a parallelogram?

ANSWER Yes — Test 2.

3. The diagonals bisect each other. Is  $ABCD$  a parallelogram?

ANSWER Yes — Test 3.

4. Only  $AB \parallel CD$  is known. What is  $ABCD$ ?

ANSWER A trapezium.

5.  $\angle A = \angle C$  and  $\angle B = \angle D$ . Is  $ABCD$  a parallelogram?

ANSWER Yes.

6.  $AC = BD$ . Is  $ABCD$  a parallelogram?

ANSWER Not necessarily.

7.  $AB = BC = CD = DA$ . Is  $ABCD$  a parallelogram?

ANSWER Yes — a rhombus, which is a parallelogram.

Medium · the standard question

7 PROBLEMS

1.  $AO = OC = 5$ ,  $BO = 4$ ,  $OD = 6$ . Is  $ABCD$  a parallelogram?

ANSWER No — only one diagonal is bisected.

2.  $AB = 7$ ,  $CD = 7$ ,  $BC \parallel AD$ . Is  $ABCD$  a parallelogram?

ANSWER Not necessarily — an isosceles trapezium fits.

3.  $M$  is the midpoint of both  $AC$  and  $BD$ . Prove  $ABCD$  is a parallelogram.

ANSWER The diagonals bisect each other — Test 3.

4.  $\angle A = 80^\circ$  and  $\angle C = 80^\circ$ ,  $\angle B = 100^\circ$ . Find  $\angle D$  and decide.

ANSWER  $\angle D = 100^\circ$ ; opposite angles equal, so it is a parallelogram.

5.  $AB = 6$ ,  $BC = 9$ ,  $CD = 6$ ,  $DA = 9$ . Is  $ABCD$  a parallelogram?

ANSWER Yes — Test 2.

6.  $AB = 6$ ,  $BC = 9$ ,  $CD = 9$ ,  $DA = 6$ . Is  $ABCD$  a parallelogram?

ANSWER No — the equal sides are adjacent; this is a kite.

7. Which test proves a rhombus is a parallelogram?

ANSWER Test 2 — both pairs of opposite sides are equal.

### Hard · stretch and reason

7 PROBLEMS

1. In  $\triangle ABC$ ,  $M$  is the midpoint of  $BC$ . Point  $D$  is on ray  $AM$  with  $AM = MD$ . Prove  $ABDC$  is a parallelogram.

ANSWER  $M$  is the midpoint of both  $BC$  and  $AD$ , so the diagonals bisect each other — Test 3.

2.  $K$  and  $L$  are the midpoints of  $AB$  and  $CD$  in parallelogram  $ABCD$ . Prove  $AKCL$  is a parallelogram.

ANSWER  $AK = \frac{1}{2} AB = \frac{1}{2} CD = CL$  and  $AK \parallel CL$  — Test 1.

3. The diagonals of  $ABCD$  meet at  $O$  and  $\triangle AOB \cong \triangle COD$ . Must  $ABCD$  be a parallelogram?

ANSWER Yes if the congruence pairs  $AO$  with  $CO$  and  $BO$  with  $DO$  — then the diagonals bisect each other.

4. Prove that if the opposite angles of a quadrilateral are equal, it is a parallelogram.

ANSWER  $2\angle A + 2\angle B = 360^\circ$ , so  $\angle A + \angle B = 180^\circ$  — co-interior angles make the sides parallel.

5. Give a quadrilateral with  $AB = CD$ ,  $AD = BC$  that is not a parallelogram, or explain why none exists.

ANSWER None exists — Test 2 proves every such quadrilateral is a parallelogram (for a simple, non-crossed quadrilateral).

6.  $E$  is the midpoint of  $AB$  and  $F$  of  $DC$  in parallelogram  $ABCD$ . Prove  $AEFD$  is a parallelogram.

ANSWER  $AE = \frac{1}{2} AB = \frac{1}{2} DC = DF$  and  $AE \parallel DF$  — Test 1.

7. In parallelogram  $ABCD$ , points  $E$  and  $F$  lie on the diagonal  $AC$  with  $AE = FC$ . Prove  $BEDF$  is a parallelogram.

ANSWER The diagonals  $BD$  and  $EF$  share the midpoint  $O$  of  $AC$ , so they bisect each other — Test 3.

## 07 Homework

### Homework — 5 problems

Geometry 8, Tema 3, pp. 11–13. Name the test you used in each proof.

1.  $AB = 9$ ,  $CD = 9$  and  $AB \parallel CD$ . Prove  $ABCD$  is a parallelogram and name the test.
2.  $AO = 4$ ,  $OC = 4$ ,  $BO = 7$ ,  $OD = 7$ . Prove  $ABCD$  is a parallelogram.
3. Give an example of a quadrilateral with equal diagonals that is not a parallelogram.
4. In  $\triangle ABC$ ,  $M$  is the midpoint of  $AC$  and  $BM$  is extended to  $D$  with  $BM = MD$ . Prove  $ABCD$  is a parallelogram.
5. Write the three tests in your own words, each with a small labelled sketch.